

subject: moving up from mechanics to electronics – the plan, and what to tell parents

Hi all,

This note helps anyone guiding a family's decision to move a child from **Level 1 mechanics** to **Level 2 electronics**. It explains – in plain words – **why the mechanics matters for the electronics**, when a child can skip straight ahead, how to prepare, and **exactly what to tell parents**.

what each level teaches

level 1 · now

mechanics

The **simple machines** every bigger machine is built from – **levers, pulleys, wheels & axles, gears**. Children build real machines (see-saw, crane, tow truck...) and discover how a machine makes **heavy work easier**.

level 2 · next

electronics

How **electricity does work** – **circuits, motors and sensors**. Children build machines that **move, light up and sense** (railway barrier, wind turbine, sensor crane...), run by switches and sensors.

why a child needs the mechanics to build the electronics

Here's the part that's easy to picture. In electronics, a **motor** does just one thing – it **spins**. Spinning on its own can't lift a crane, drive a truck, or open a bridge. A **simple machine** is what turns that spin into the actual job – and that is exactly what mechanics teaches. The motor is the muscle; the **pulley, wheel, gear or lever** is what makes it useful:

the motor just spins – the simple machine turns that spin into...	...which is how they build
a PULLEY → lifting up & down	the sensor crane · the elevator
a WHEEL & AXLE → rolling & driving	the carrier truck · the bots
GEARS → more force, or a new speed	the wind turbine · geared machines
a LEVER → a big up-and-down swing	the railway barrier · the drawbridge · the robotic arm

So a child who already understands pulleys, wheels, gears and levers looks at a motorised crane and **gets it** – the motor turns, the pulley lifts. A child who hasn't just sees a mystery box. That's why mechanics first isn't busywork: **it's what makes the electronics make sense**.

it also builds the skills and the thinking

Two more reasons, beyond the machines themselves. First, the **skills are the same four** in both levels – read a build card, follow a manual, run a **predict → test → find** experiment, troubleshoot, explain. Mechanics is where a child first makes them secure, so electronics isn't teaching *how to build and investigate* and *electricity* at once. Second, mechanics is **concrete** – you can see effort, load and motion – while electricity is **invisible**; meeting the ideas hands-on first makes the abstract far easier (the well-established concrete → pictorial → abstract path).

why we teach every skill – nothing is assumed

Every skill is **taught through doing** – none is assumed to be “natural” or already there. So **a child is never “behind”**: if the skills aren't secure yet, the learning time simply **isn't finished**, and more mechanics is what finishes it. A foundation we build, not a talent we test.

when a child can start electronics directly

Age isn't the gate – the skills and concepts are. If a child has already done a similar hands-on building or robotics programme – **at school, elsewhere, or with Openhouse in the past** – and already understands the simple machines and shows the four skills (they build confidently, see how a machine works, solve problems, and can explain it), we **don't repeat mechanics**. We run the one-hour assessment below to **confirm** they're ready, and start them in electronics.

preparing to run it · when to run it

prepare (~10 min)

- **pre-build the tow-truck chassis** so the hour is about the winch, wheels and tests
- lay out: the **pulley-card loads** (0.5 / 1 / 1.5 kg), a **box**, a **rope**, a **cart** (or the truck's own wheels), the **two experiment cards** and a **pencil** – a **spring scale** too if your centre has one
- print the child's booklet

when to run it

- when a child has **finished (or nearly finished) mechanics** and a family is considering electronics
- when a **new child joins** who has done a similar programme elsewhere and may skip ahead
- any calm **1-hour slot, one teacher : one child** – not inside a busy group session

what to tell parents – in plain words

The four questions parents ask, with a clear answer for each – say them, or send them as-is:

- " **Why does mechanics come first?** In electronics your child builds machines like a motorised crane or a delivery truck. The motor only spins – it's the **pulley** that turns that spin into lifting, and the **wheels** that turn it into driving. Knowing those simple machines first means your child understands the whole machine, not just the wiring.
- " **Is my child behind?** Not at all. Everything here is taught, never assumed. If we suggest a little more mechanics, it only means the foundation isn't finished yet – that's what completes it.
- " **My child has done robotics before.** Wonderful – they may go straight to electronics. We do a short, friendly one-hour check to be sure the simple-machine ideas and the skills are there, then move them up.
- " **What will they actually learn?** Mechanics: how simple machines make heavy work easier. Electronics: how circuits, motors and sensors make things move, light up and sense.

the assessment – in brief

A warm ~1-hour check, one teacher + one child, around the child's favourite mixed machine, the **tow truck**. It's **one test for ages 5-12** – you flex the bar (younger children speak and you scribe; older children write, measure and explain in full). It should feel like a mission, not a test. Three parts:

1

this note

the why & what to tell parents

2

the child's booklet

four challenges – they keep it, show it at home

3

the marking sheet

watch & mark – teacher only, not for parents

Warm up with a chat, then the child leads four challenges – **build it · investigate it · apply it · be the teacher** – each proving one skill. Mark each as it happens, reflect, then decide: **ready for electronics**, or **a little more mechanics first**. Everything is guided on the marking sheet.

ps – the marking sheet is for the teacher, never the parent. The child keeps their booklet and shows it proudly at home. Every child leaves having **built, tested, applied and taught** – a good day either way.

Thanks for making this a celebration,
– the openhouse robotics team

⚠ for the teacher only – keep this; do not send it home.

■ at a glance

A friendly ~1-hour session, one teacher + one child, around the **tow truck** (wheel & axle + pulley). The child works through **four challenges** from their booklet, mostly on their own. Your job: **warm them up, set up, then watch and mark** – each mark below is tied to **exactly what the child does in that challenge**. It's a **readiness** check, not a grade.

■ one test, ages 5–12

you adapt, not the paper

The same booklet suits both bands – **you** flex the bar. **Ages 5–8:** read the prompts aloud and scribe their spoken answers; “ready” = clear spoken reasoning + one real-life example. **Ages 8–12:** expect written answers, effort **measured** with the spring scale, precise reasons, and **more than one** example – the ★ notes below mark this higher bar. Mark each child against **their own age**.

■ warm-up – get them talking

🕒 3 min

Chat first – it settles the child and shows you where they're starting from. Ask:

- “what's the best machine you've built with us? what did it do?” (*warms them up · shows recall*)
- “can you point to something here that **rolls**? something that **lifts**?” (*shows they link parts to jobs*)
- “what does a **pulley** help us do?” (*a quick read of prior understanding*)

■ set up the station

🕒 5 min

- the child's **tow-truck kit + manual** – **pre-build the chassis** so the hour is about the winch, wheels and tests
- the **loads from the pulley card** – 0.5 kg, 1 kg and 1.5 kg (a **spring scale** too, if your centre uses one, to measure the effort – otherwise the child just feels it)
- a **box**, a **rope**, and a **cart** (the tow truck's own wheels work) – for the wheel & axle test
- the two experiment cards (**pulley · wheel & axle**) and a **pencil** – the child reads, records and writes on their booklet

■ how to run it

- Read the one-line brief at the top of each challenge, then **step back**. One guiding question only if truly stuck – **never fix it for them**.
- Watch each challenge and mark it on page 2 **as it happens** (the marks below describe exactly what to look for).
- Keep it light and encouraging – a chance to shine, not a test to pass.

Each mark below is **exactly what the child is doing in that challenge** – watch for it and circle one. “ready ★” is the milestone for that skill.



challenge 1 · build it building & making

🕒 15 min

the child is: building the truck + winch from the manual, then **labelling the parts on the diagram** (matching pulley · wheel & axle · hook) and marking how the winch lifts (handle → load goes up · the pulley does the lifting).

not yet needs help to build, or can't match the parts to the diagram

developing builds it working AND labels all three parts correctly

ready ★ all that AND fixes a slip / loose part on their own, and knows the **pulley (winch)** is the part that does the lifting

mark: ☐ not yet ☐ developing ☐ ready



challenge 2 · investigate it observing & understanding

🕒 18 min

the child is: recording effort for 0.5 / 1 / 1.5 kg (by hand vs winch) in the data table, testing the wheels (drag vs roll), then answering **why** simple machines make work easier.

not yet does the tests but can't read the pattern or fill the table sensibly

developing records the data AND states both patterns – heavier = more effort; the machine needs less effort

ready ★ all that AND explains the CONCEPT in their own words – a simple machine lets you use less effort to do the same work (not “makes the load lighter”)

mark: ☐ not yet ☐ developing ☐ ready



challenge 3 · apply it problem solving

🕒 15 min

the child is: naming the simple machine in three machines they did NOT build (flagpole → pulley · bicycle → wheel & axle · scissors → lever), then solving two real problems – raise a flag from the bottom, and move heavy sacks across a yard.

not yet can't spot the machines in the everyday objects, or picks the wrong one for the problems

developing identifies the three machines correctly, and picks the pulley (flag) and wheels/cart (sacks) with a reason

ready ★ all that AND explains HOW each makes the work easier (pulley changes the pull's direction · wheels cut the drag) – real transfer to machines they've never seen (8-12: also answers the ‘even easier’ challenge – more pulleys / share the load)

mark: ☐ not yet ☐ developing ☐ ready



challenge 4 · be the teacher presenting & explaining

🕒 12 min

the child is: reading and answering aloud the four questions printed on their sheet: (1) what a pulley is & does · (2) what a wheel & axle is + point to it · (3) WHY simple machines make work easier · (4) two real-life examples.

not yet answers few; names a part but can't say what it does

developing answers Q1 & Q2 – says what the pulley and the wheel & axle each are and do

ready ★ also answers Q3 (the WHY – less effort for the same work) and Q4 (gives real-life examples) – the clearest sign the concepts are secure

mark: ☐ not yet ☐ developing ☐ ready

reflection – ask at the end

⌚ 3 min

A short chat that lets deeper understanding show (feeds **observing & understanding** and **presenting**). Ask, and listen for the italic:

- “in your data, which needed less effort – by hand or the winch? **why is that?**” *listen for the concept – less effort for the same work – not just “the winch”*
- “if the load **doubled**, what would happen to the effort, and what would you do?” *listen for a prediction + reaching for a simple machine on purpose*
- “if you built a real **lift or crane**, which of these machines would be inside it – and why?” *a deeper transfer probe – the strongest sign the ideas are secure*

the decision

Ready to start electronics when the child is at “**ready**” on **building & making, observing & understanding, and problem solving** (with at least **developing** on presenting) – **observing & understanding** matters most, since it is the skill electronics is built on.

Otherwise: **a little more time in mechanics first**. Record it as “**not yet**” – never shown to the child as a fail. It just means the learning time isn't complete, and more mechanics is exactly what completes it.

record

child: _____ · date: _____

skills seen as “ready”: ○ building & making ○ observing & understanding ○ problem solving ○ presenting & explaining

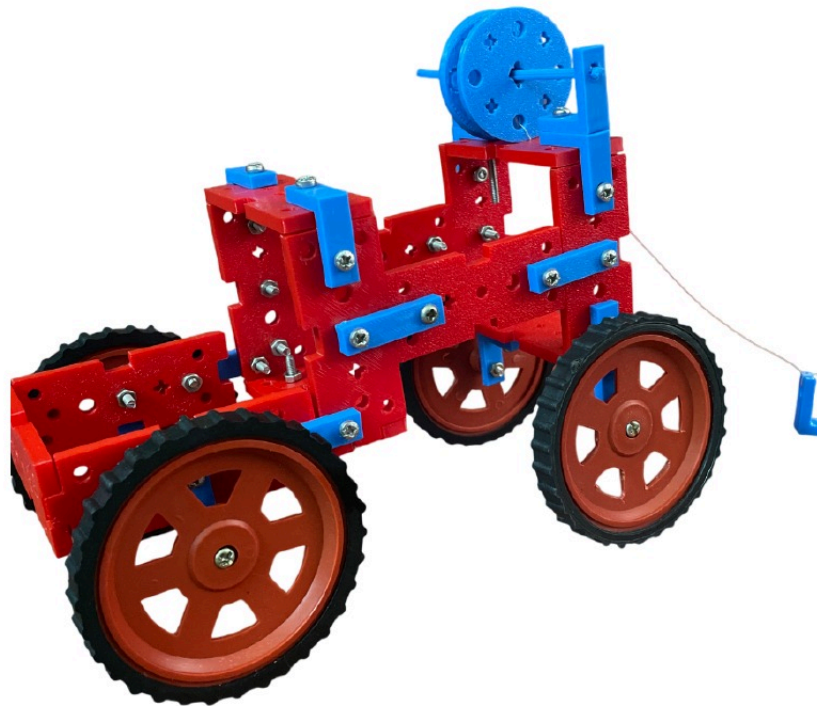
outcome: ○ ready for electronics ○ a little more mechanics first

openhouse

robotics · level-up challenge

the tow truck challenge

your mission, engineer: build it, test it, solve it, teach it! 🔧



about 1 hour

4 challenges

my name _____ today's date _____



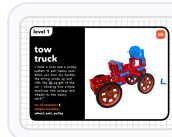
build it

building & making

🕒 15 min

■ build your tow truck and its winch

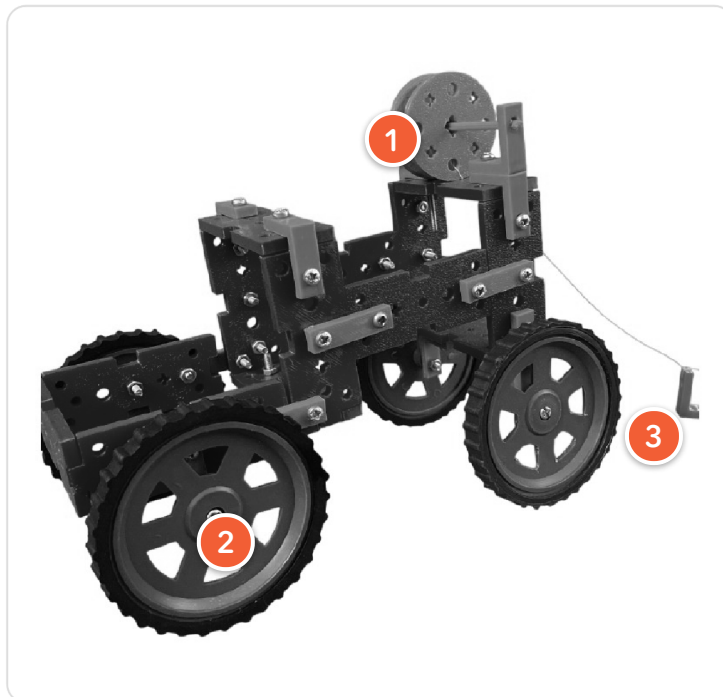
use your manual, then check both work:



manual

☐ it rolls on its wheels ☐ the winch winds up and lifts the load

■ label the parts – match each number to its part



word bank: pulley (the winch) · wheel & axle · hook

1

2

3

■ how does it work?

when you turn the handle, the string winds up and the load goes

↑ up

↓ down

the part that does the **lifting** is the

pulley

wheels



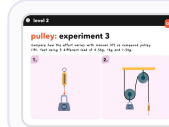
investigate it

observing & understanding

🕒 18 min

■ does the winch really make lifting easier? gather the data

lift each load **by hand**, then **with the winch**. how hard is it?



pulley card

write *easy* · *medium* · *hard*, or measure the effort with a spring scale and write the number.

load to lift	lift by hand	lift with the winch
0.5 kg		
1 kg		
1.5 kg		

■ what does your data show?

as the load gets **heavier**, the effort gets

for the same load, the **winch** needs effort than my hand.

■ now the wheels – a quick test

drag the same load along the floor, then roll it on the cart.



wheel card

which needed **less effort**? why do you think?

★ the big idea: why do simple machines like the pulley and the wheels make heavy work easier?

- ☐ they let you use less effort to do the same work ☐ they make the load lighter

explain in your own words:

.....



apply it

problem solving

🕒 15 min

you've mastered the tow truck. now use what you know on **machines you did not build**.

spot the simple machine in each everyday machine

word bank: **pulley** · **wheel & axle** · **lever**



flag on a tall pole

machine:

it helps you:



bicycle

machine:

it helps you:

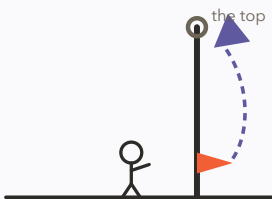


scissors

machine:

it helps you:

now solve two real problems



① A heavy flag must reach the **top of a tall pole**, but you stand at the **bottom** and can only pull **downwards**. Which machine gets it up, and how does it help?

.....

.....



② A gardener must move **heavy sacks** across a bumpy yard. Dragging them is exhausting. What would you add to make it easy, and why?

.....

.....

challenge: for problem ①, how could you make the pull even **easier** still?



be the teacher

presenting & explaining

🕒 12 min

your turn to teach! **read each question and say the answer out loud.** jot a few words on the lines so you remember.

1. what is a **pulley**? what does it do?

2. what is a **wheel & axle**? point to it on your truck.

3. **why** do simple machines make heavy work easier?

4. where else in **real life** do you see a pulley or a wheel & axle? tell two.

🎉 **mission complete, engineer!** you built it, tested it, solved it and taught it.

🌟 **now teach it to someone at home!**

who I taught _____ their high-five 🤝 _____